DATASET

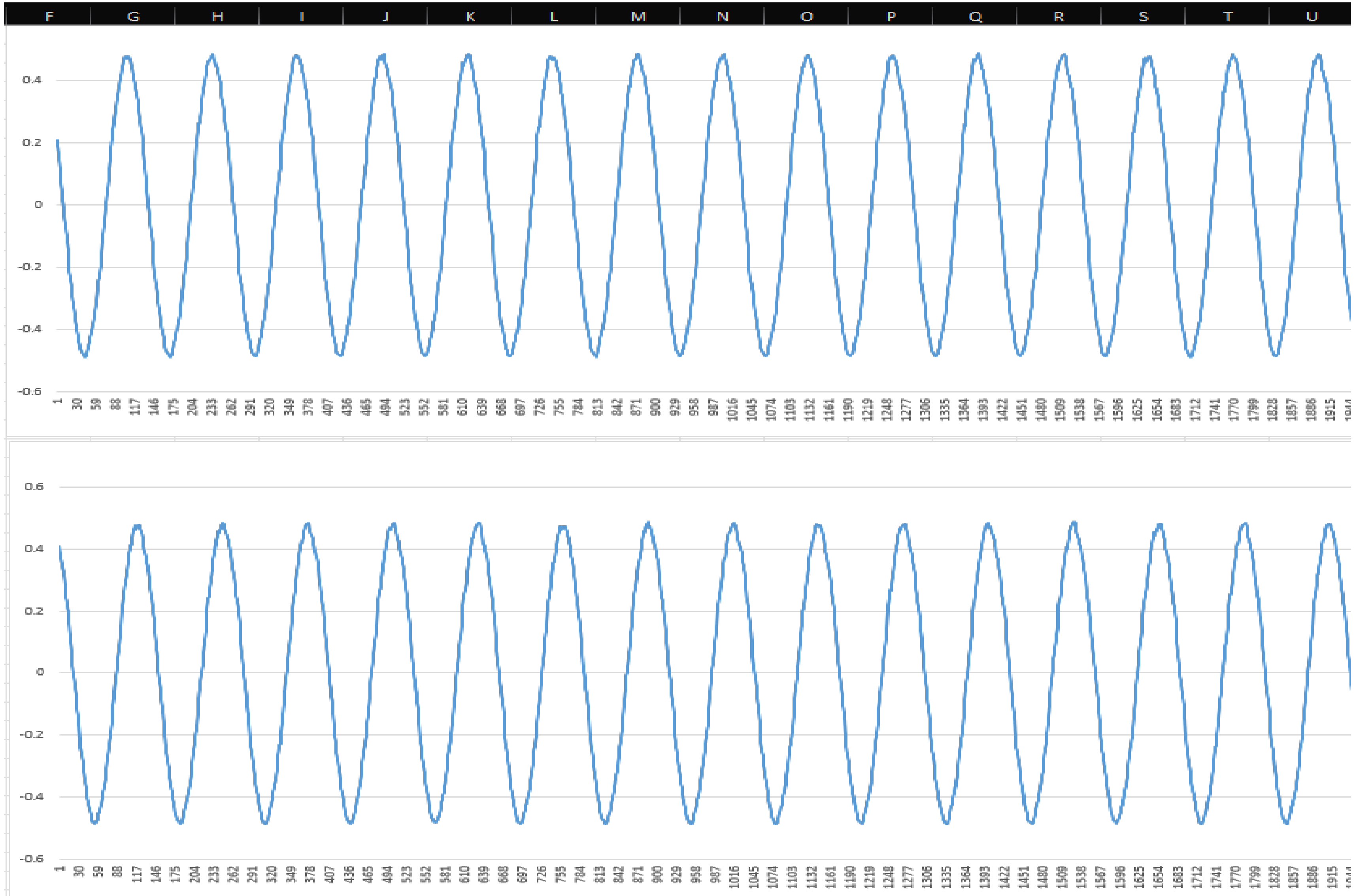
- Main Targeted Load Characteristics
  - Resistive Baseline
  - Inductive Commutator Loads
  - Non-Linear Switched Loads
  - Phase-Controlled Chopped Loads

DATASET

Halogen Lamp
Fluorescent Lamp
Electric Power Tool
Vacuum Cleaner
Switching Power Supply (SMPS)
Electronic Dimmer 60°
Electronic Dimmer 90°
Electronic Dimmer 120°
Air Compressor

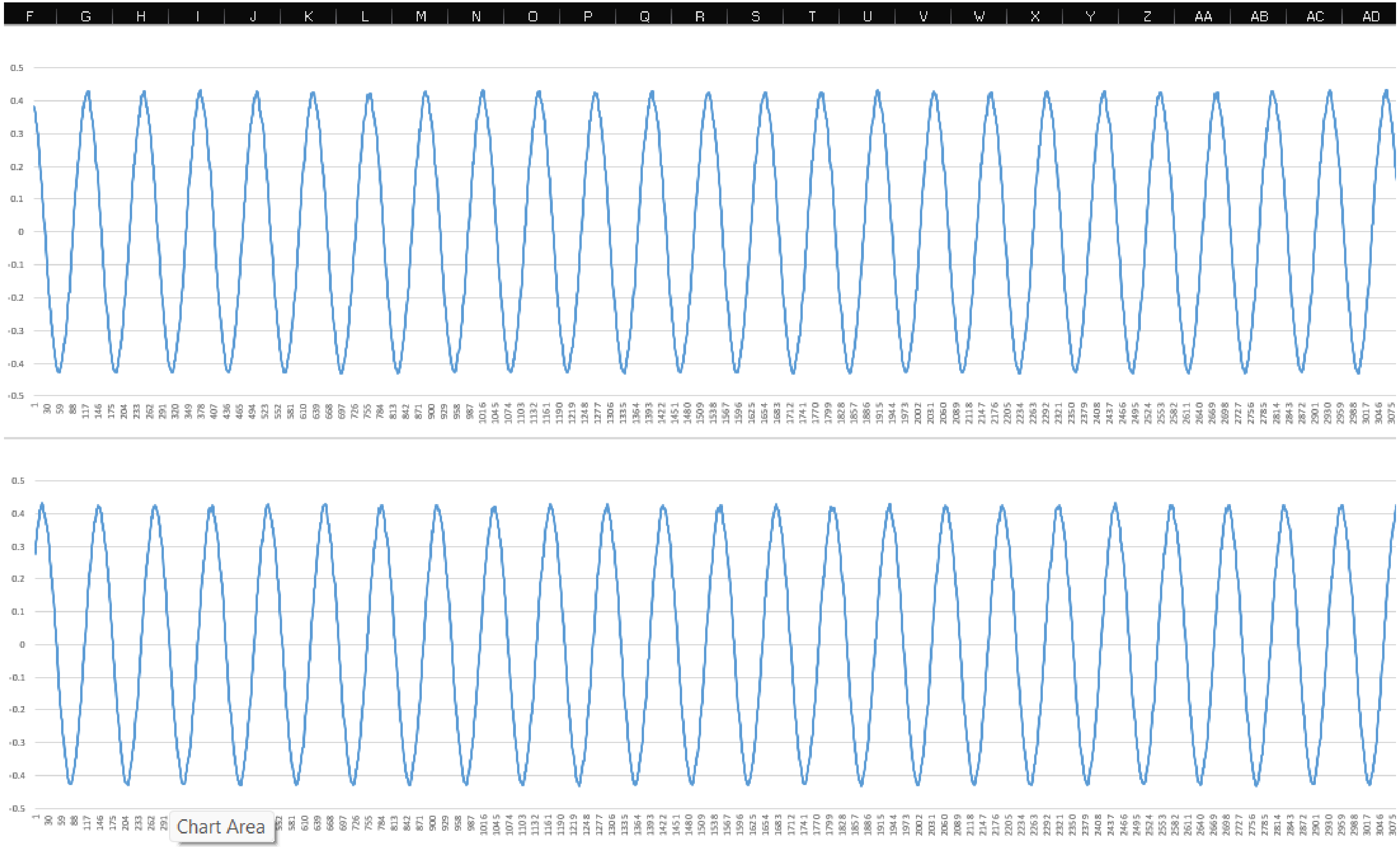
Resistive Baseline

Halogen Lamp



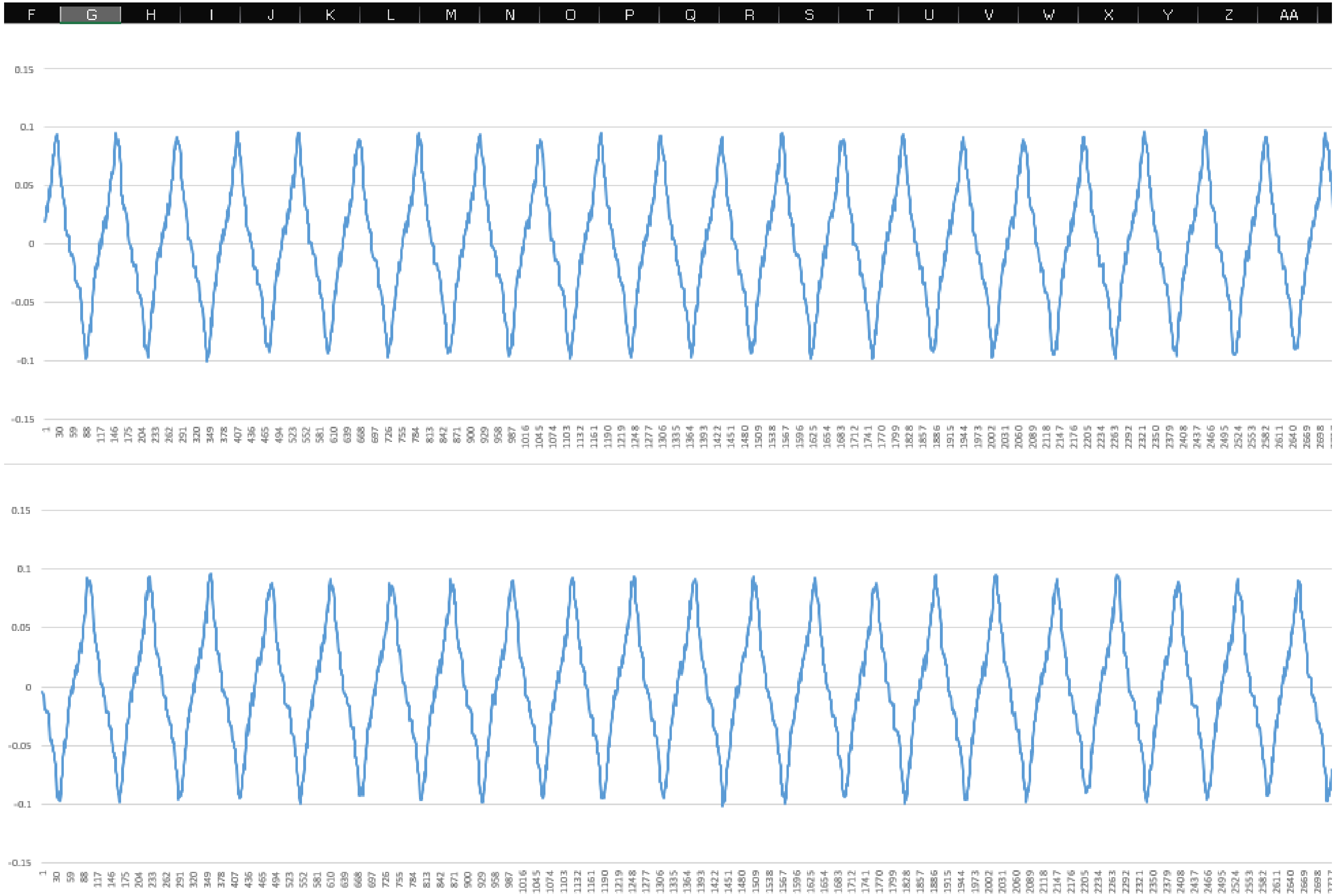
Resistive Baseline

Fluorescent Lamp

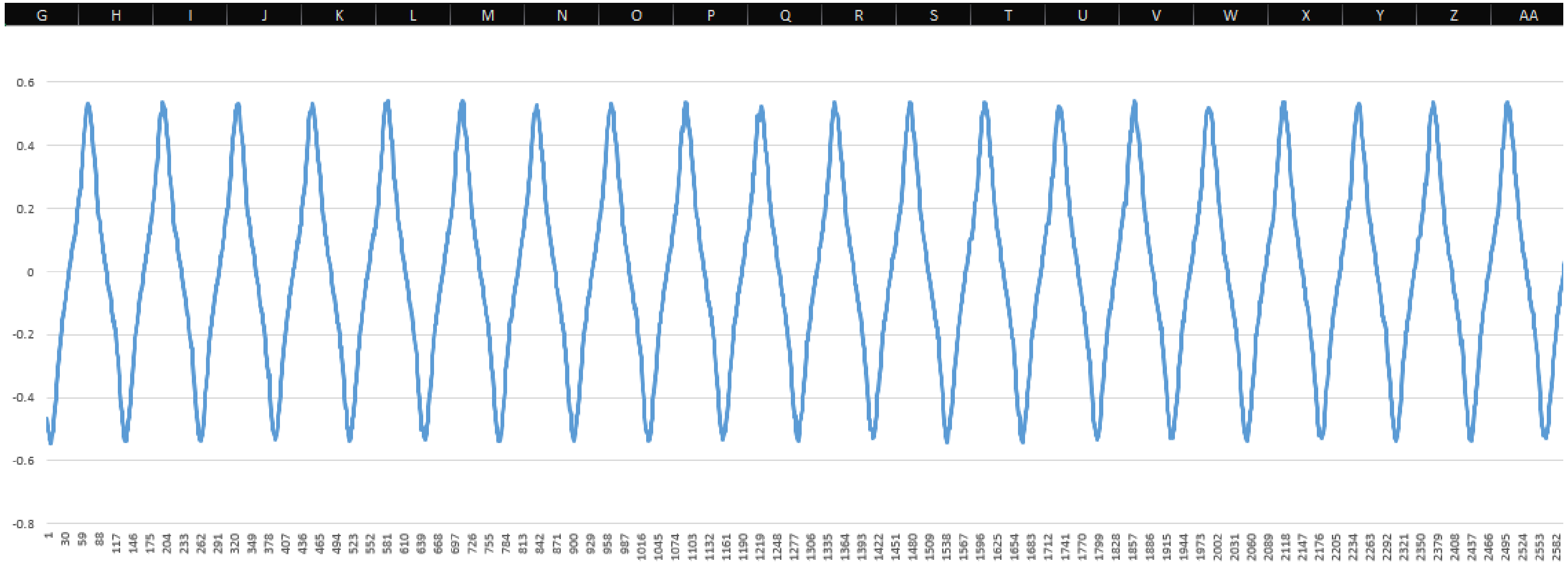


Inductive Commutator Load

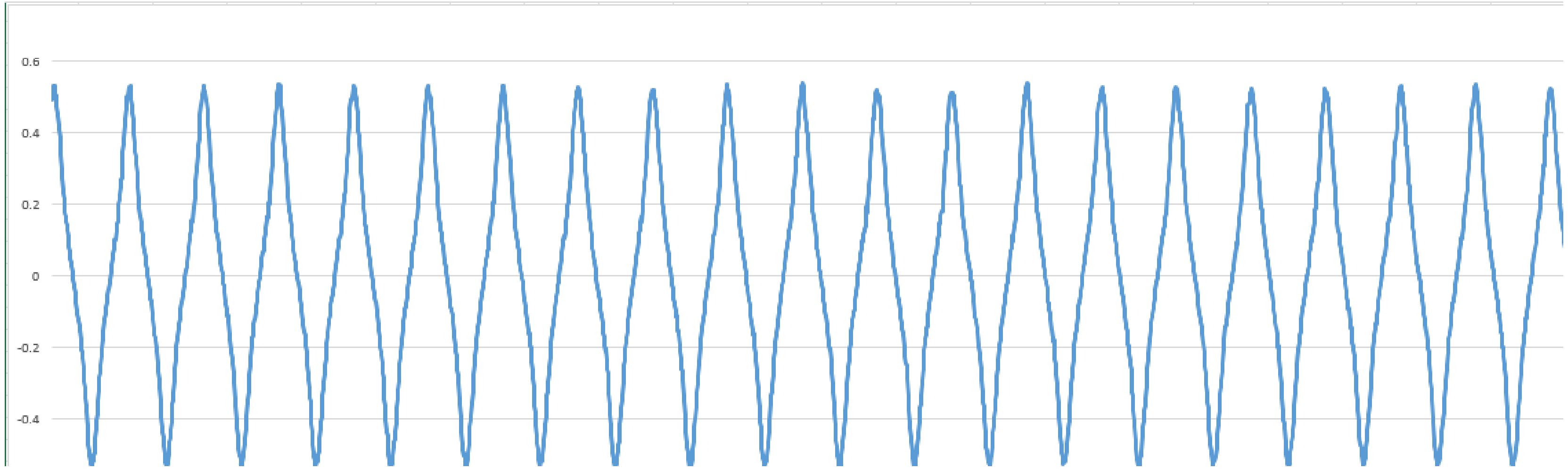
Power Tool



Inductive Commutator Load

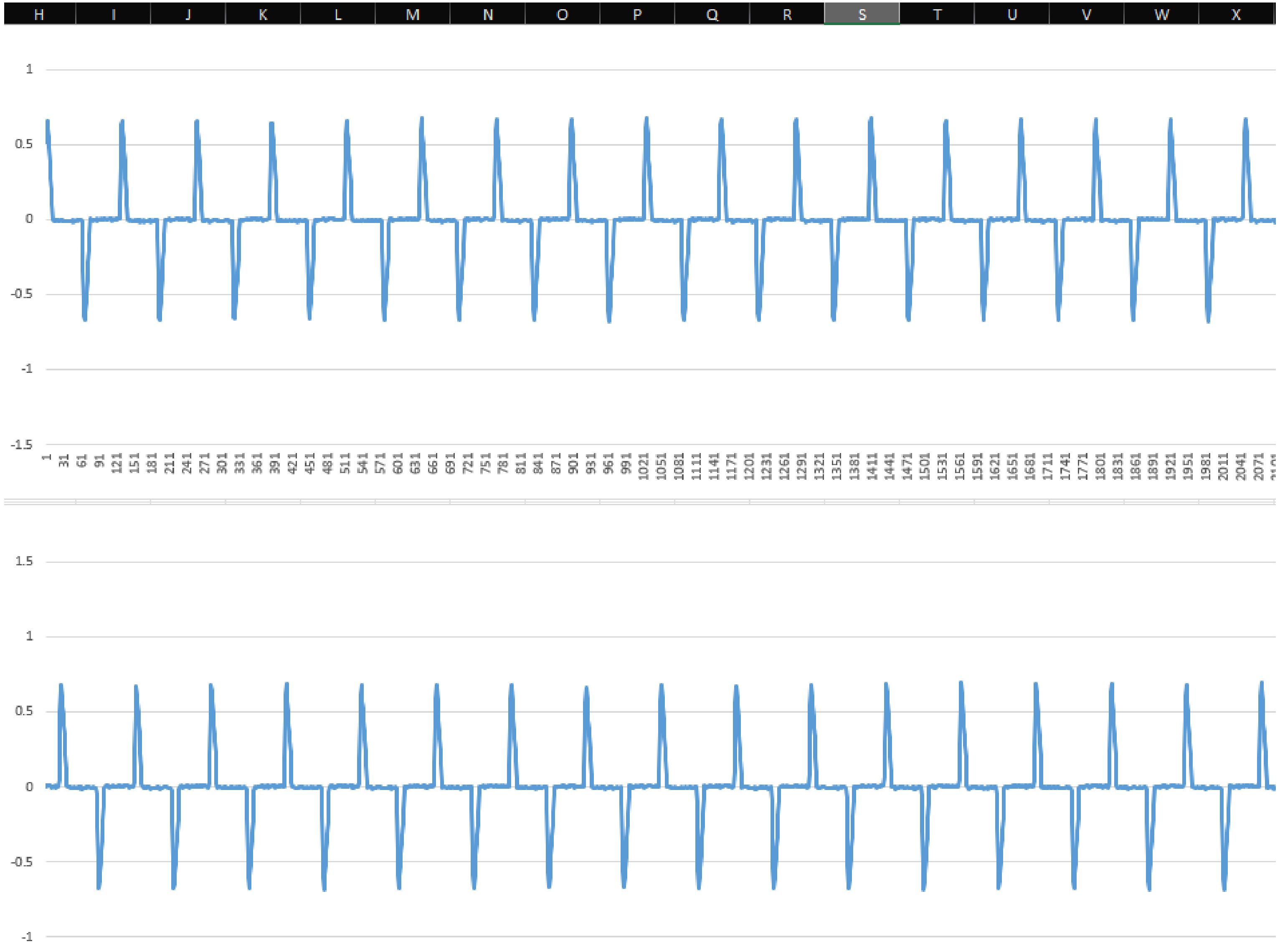


Vacuum Cleaner



Non-Linear Switched Load

Switching Power Supply (SMPS)

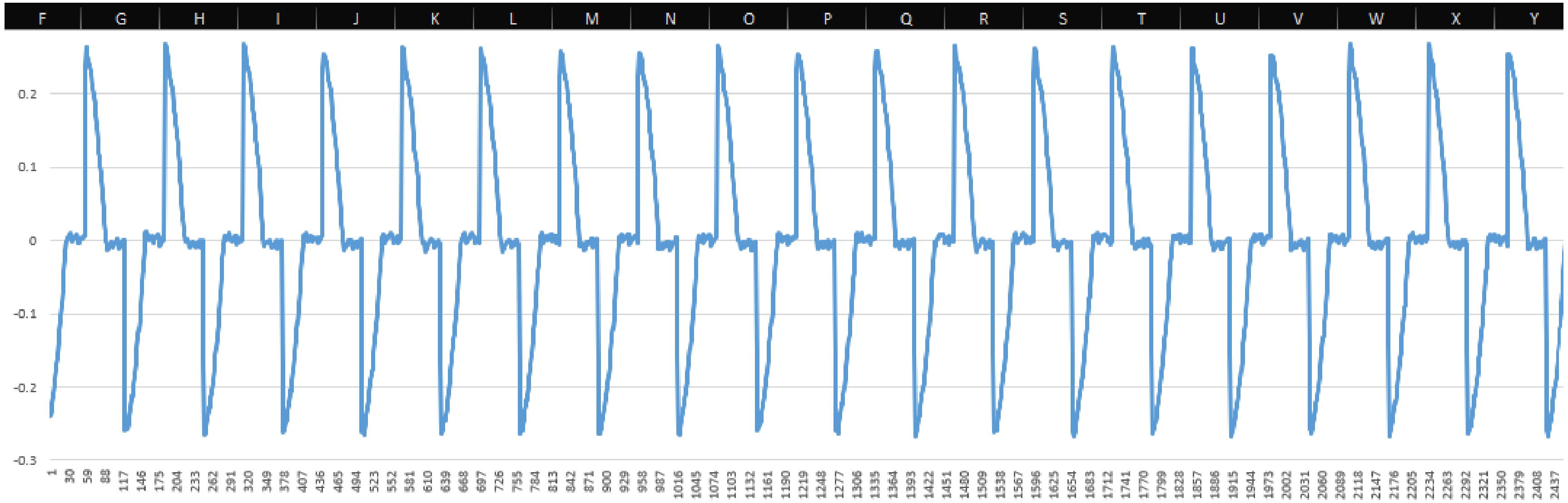


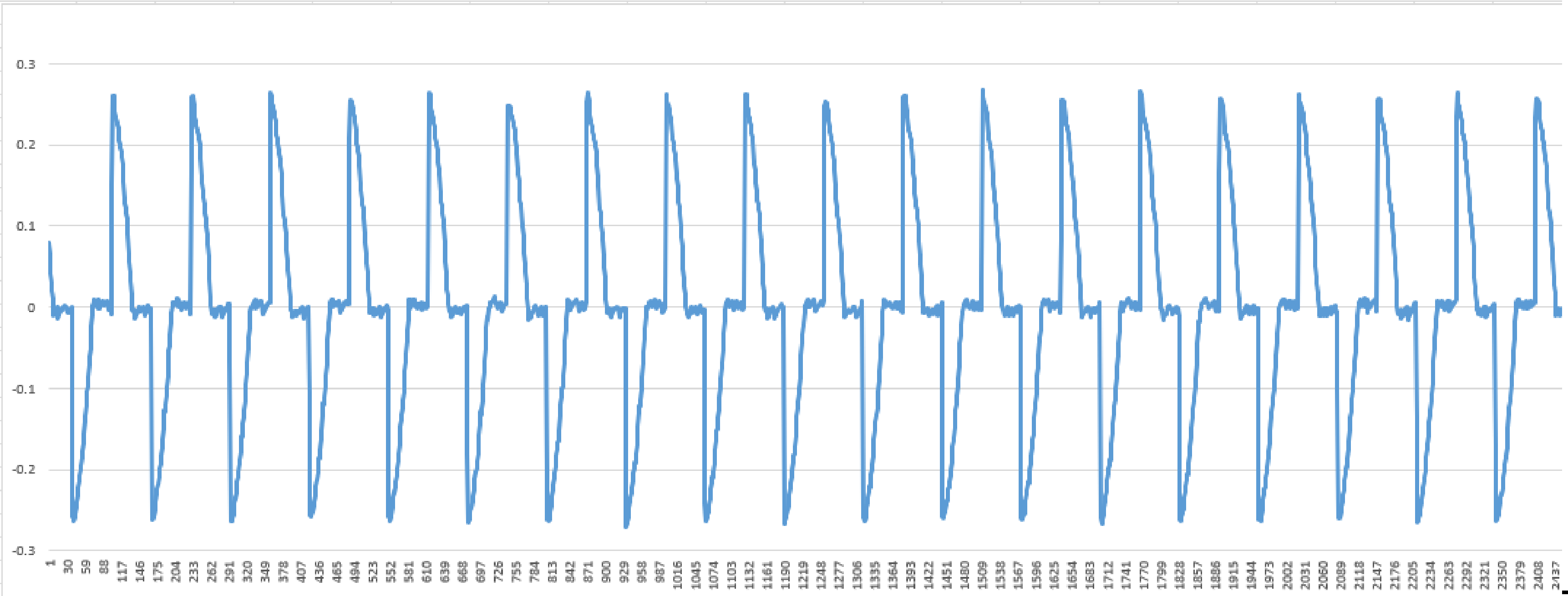
Phase-Controlled Chopped Load

Electronic Dimmer 60º

Phase-Controlled Chopped Load

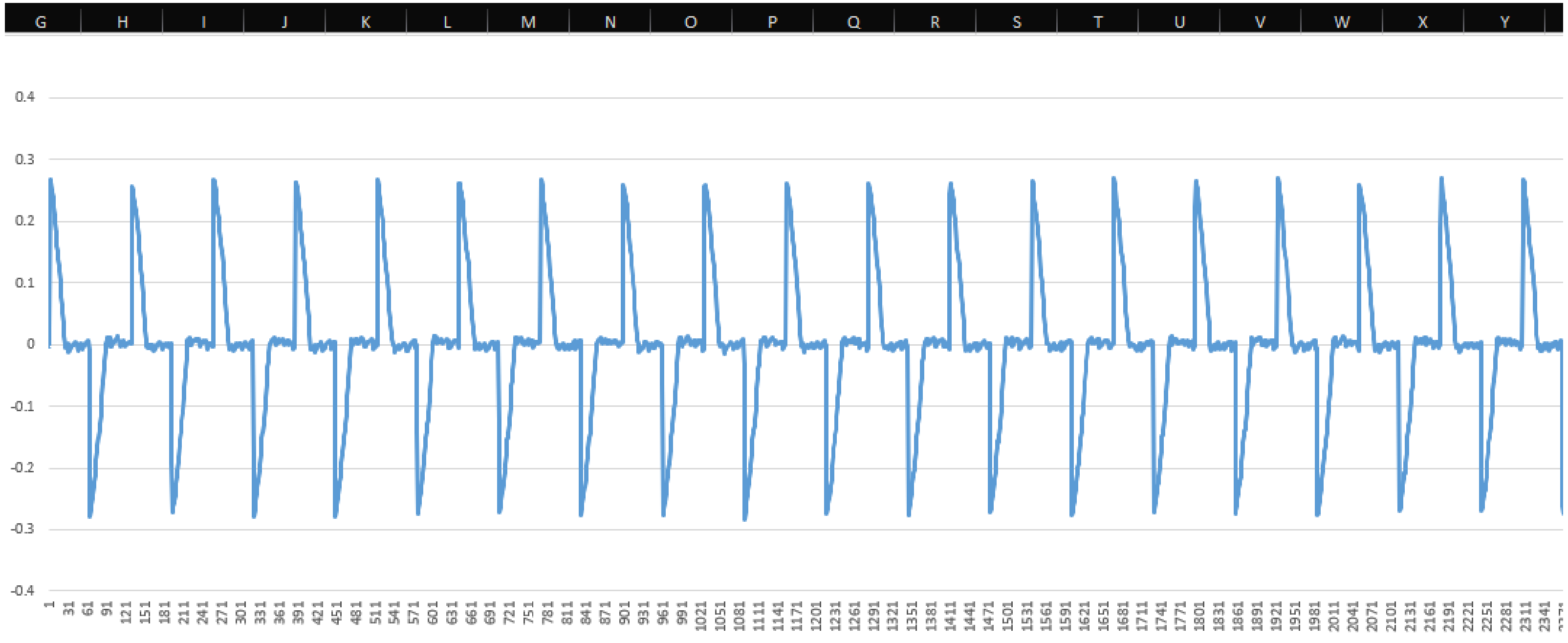
Electronic Dimmer 90°

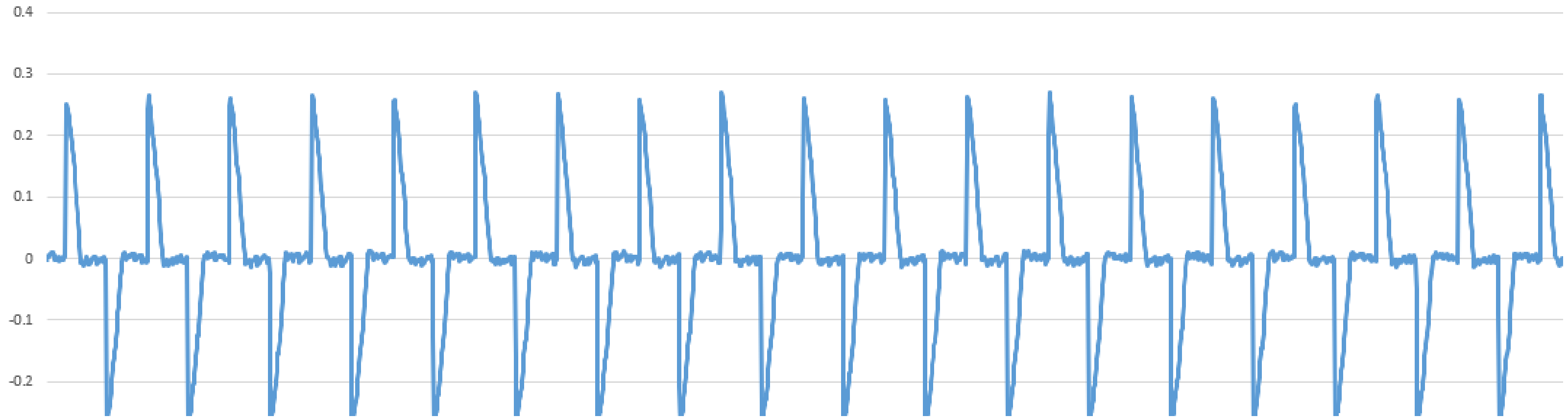




Phase-Controlled Chopped Load

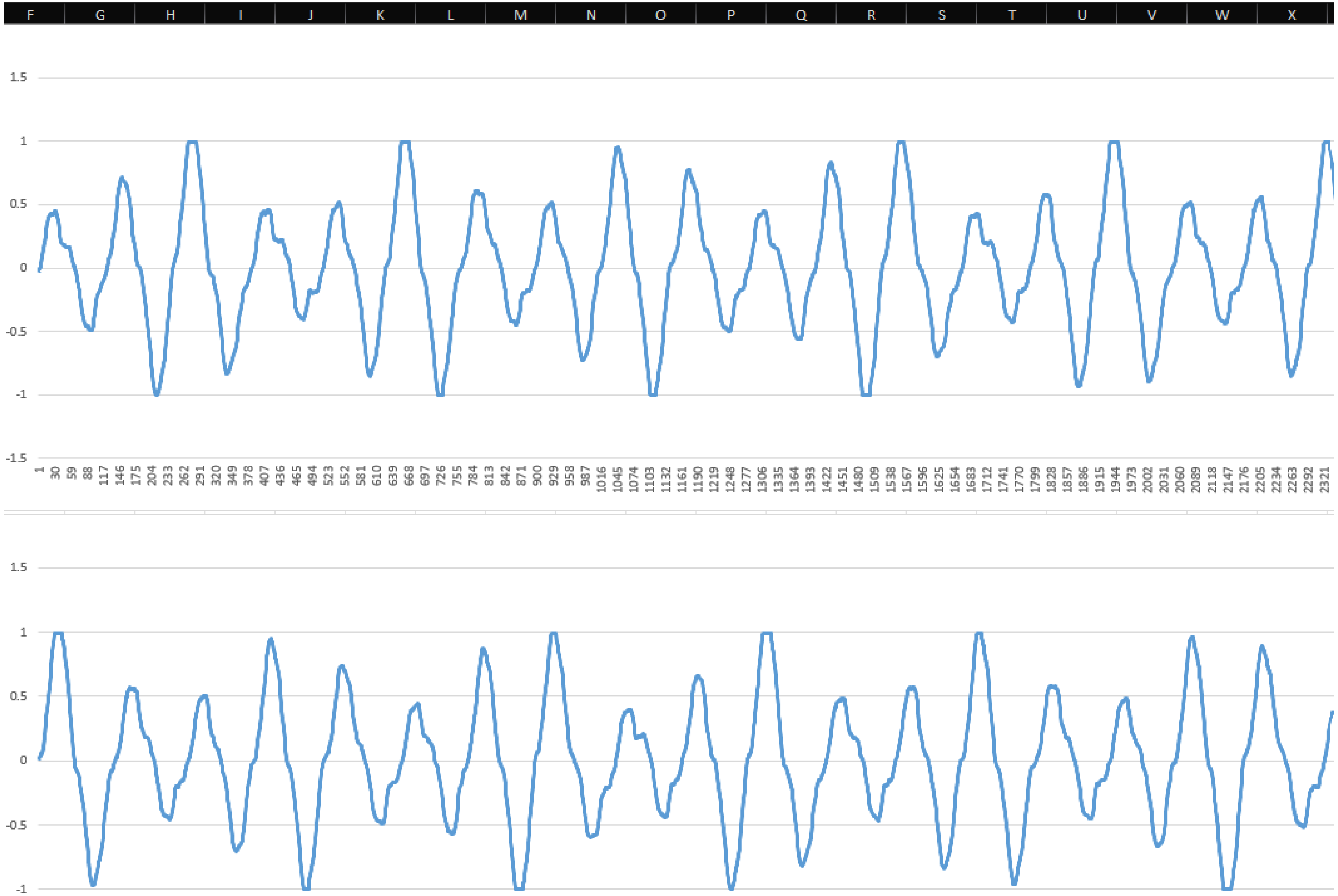
Electronic Dimmer 120°



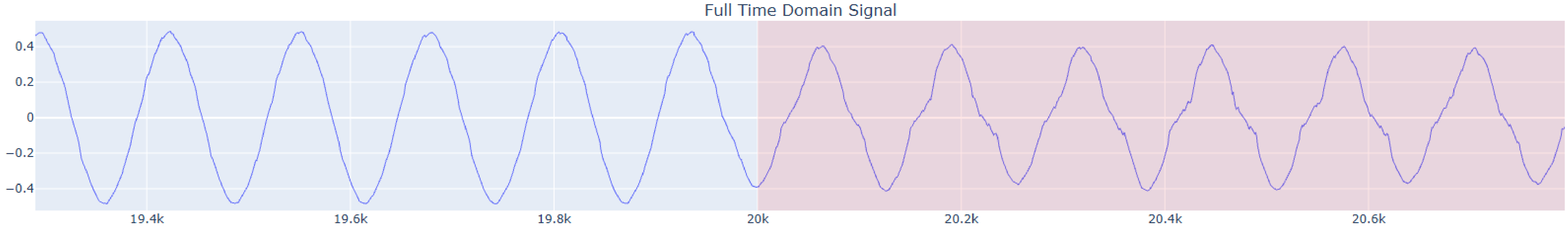


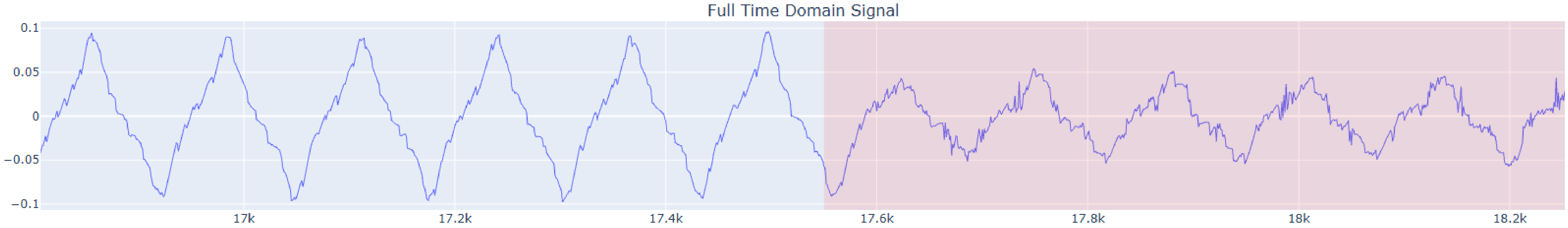
DataSet

Air Compressor

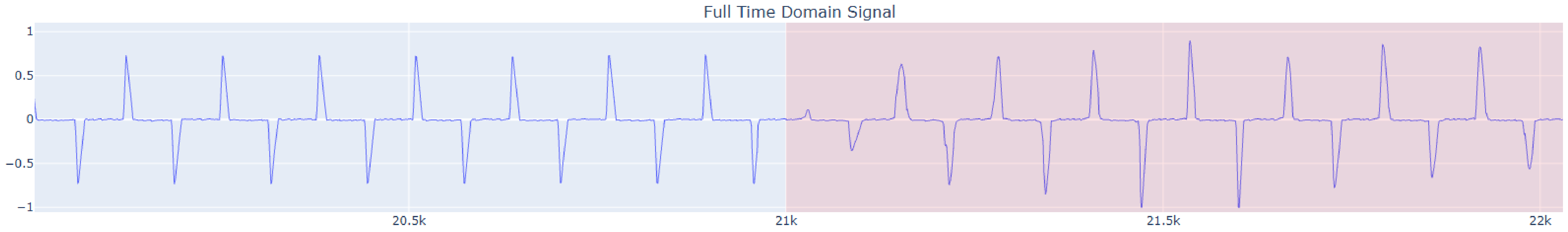


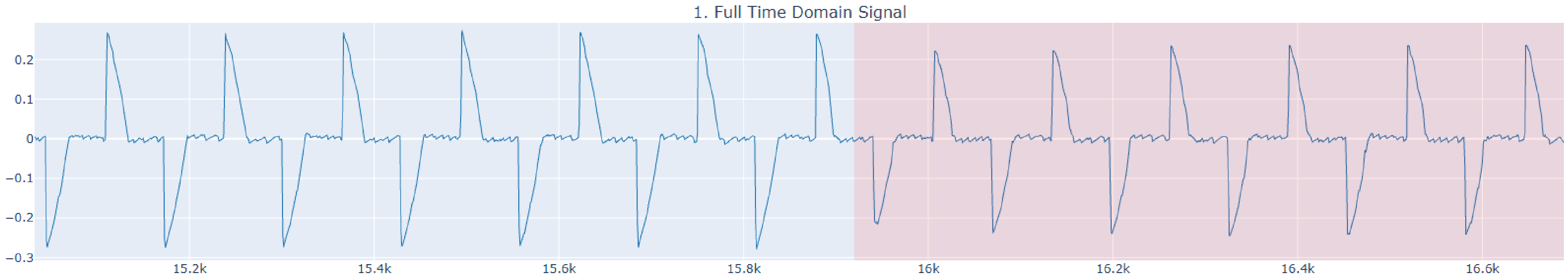
ARC Interval





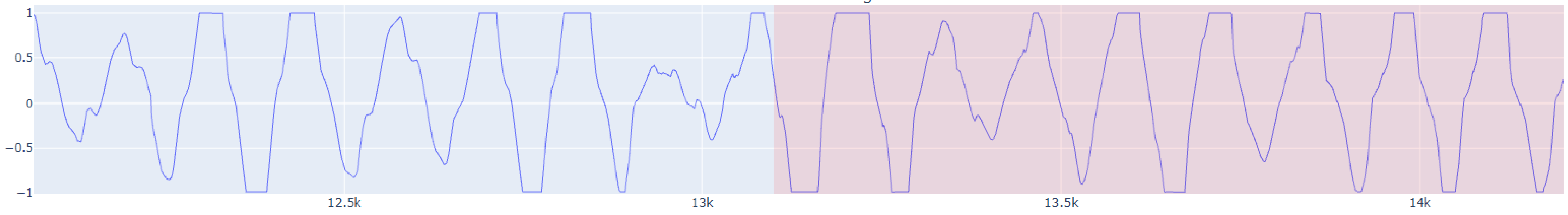
ARC Interval



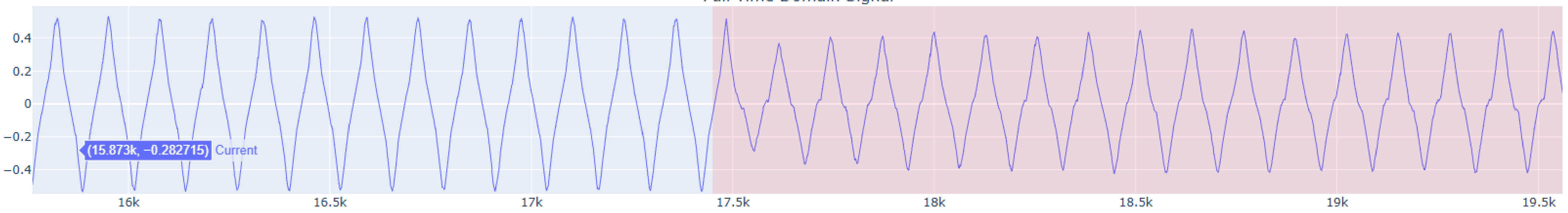


ARC Interval

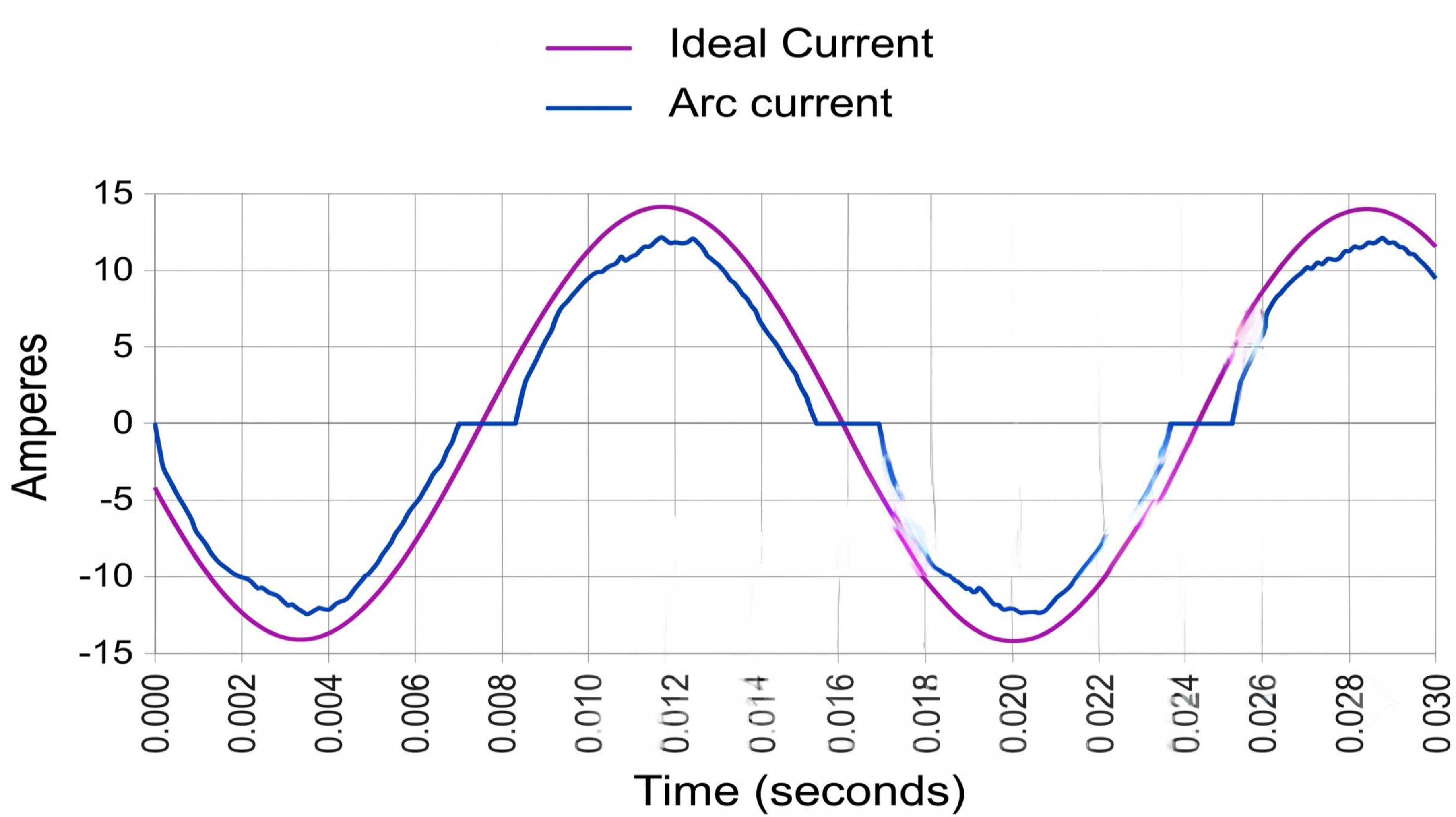
Full Time Domain Signal



Full Time Domain Signal



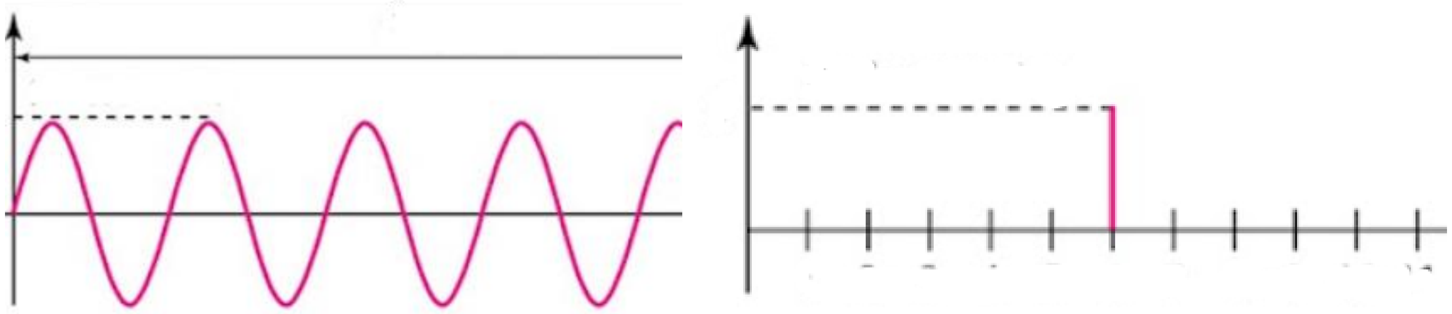
Feature Engineer



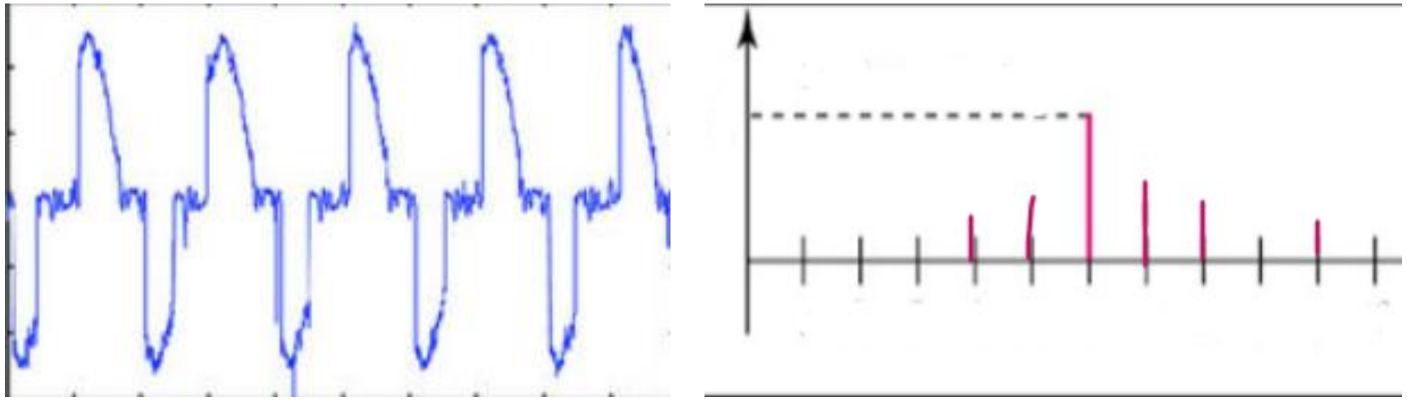
Frequency Domain

$$X[k] = \sum_{n=0}^{N-1} x\,[n] \cdot e^{-j\frac{2\pi}{N}kn}$$

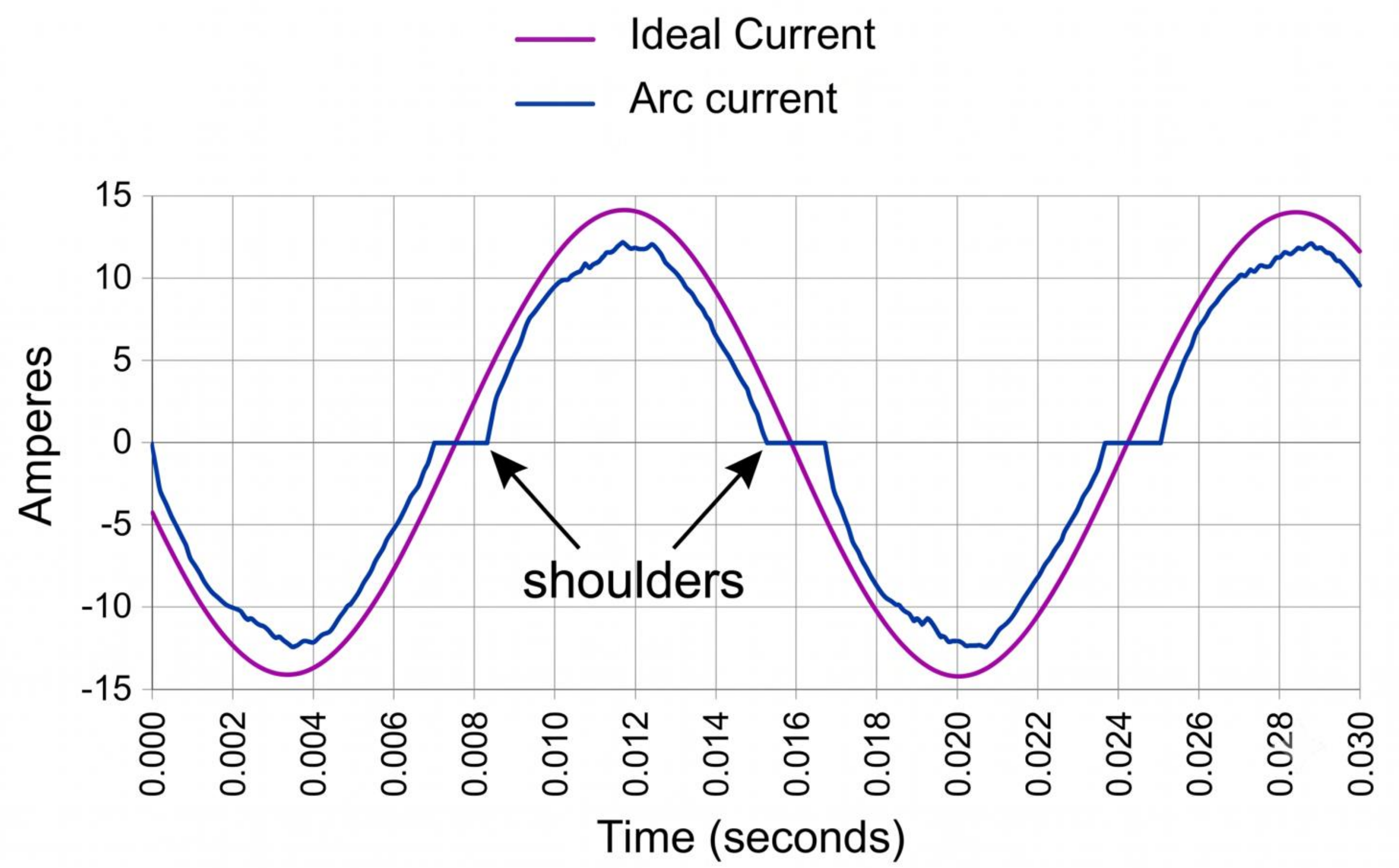
Normal



ARC

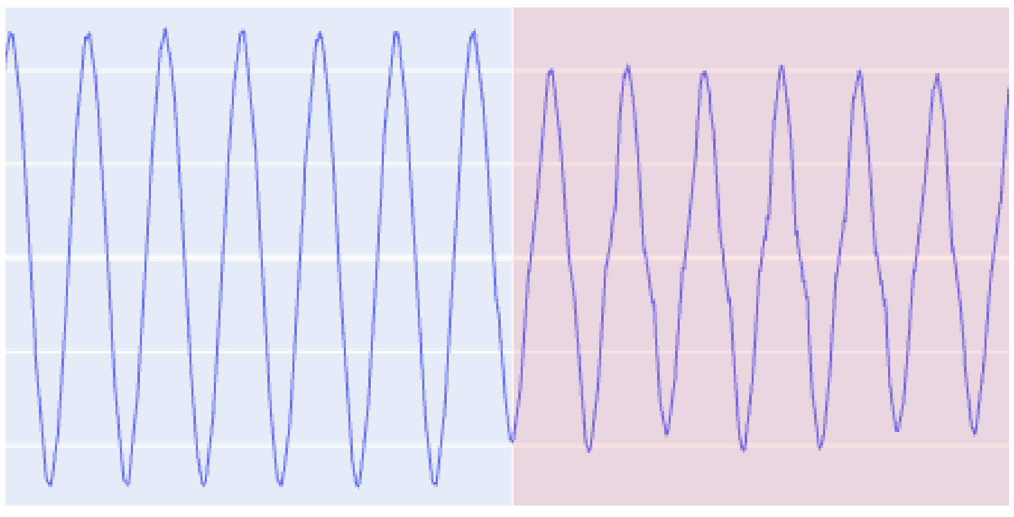


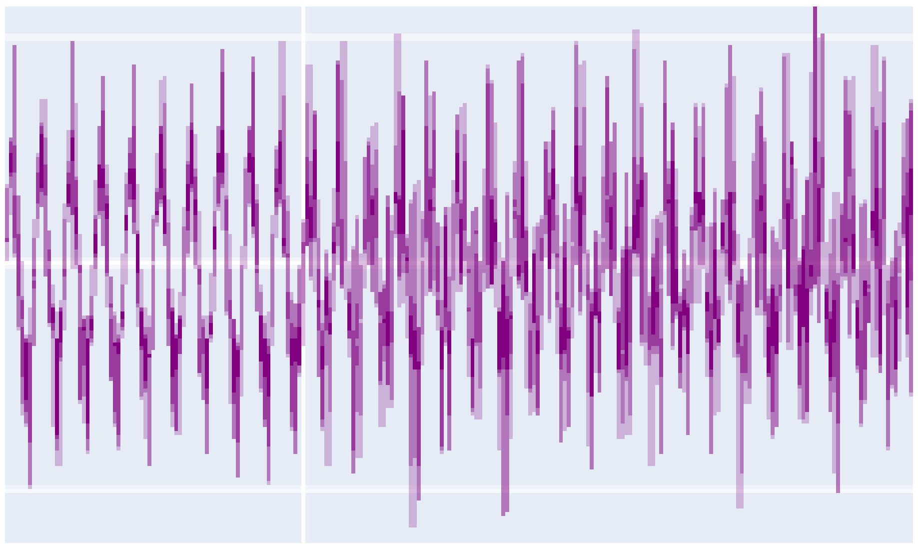
Zero-Crossing (Shoulders)



Rate of Change d/dt

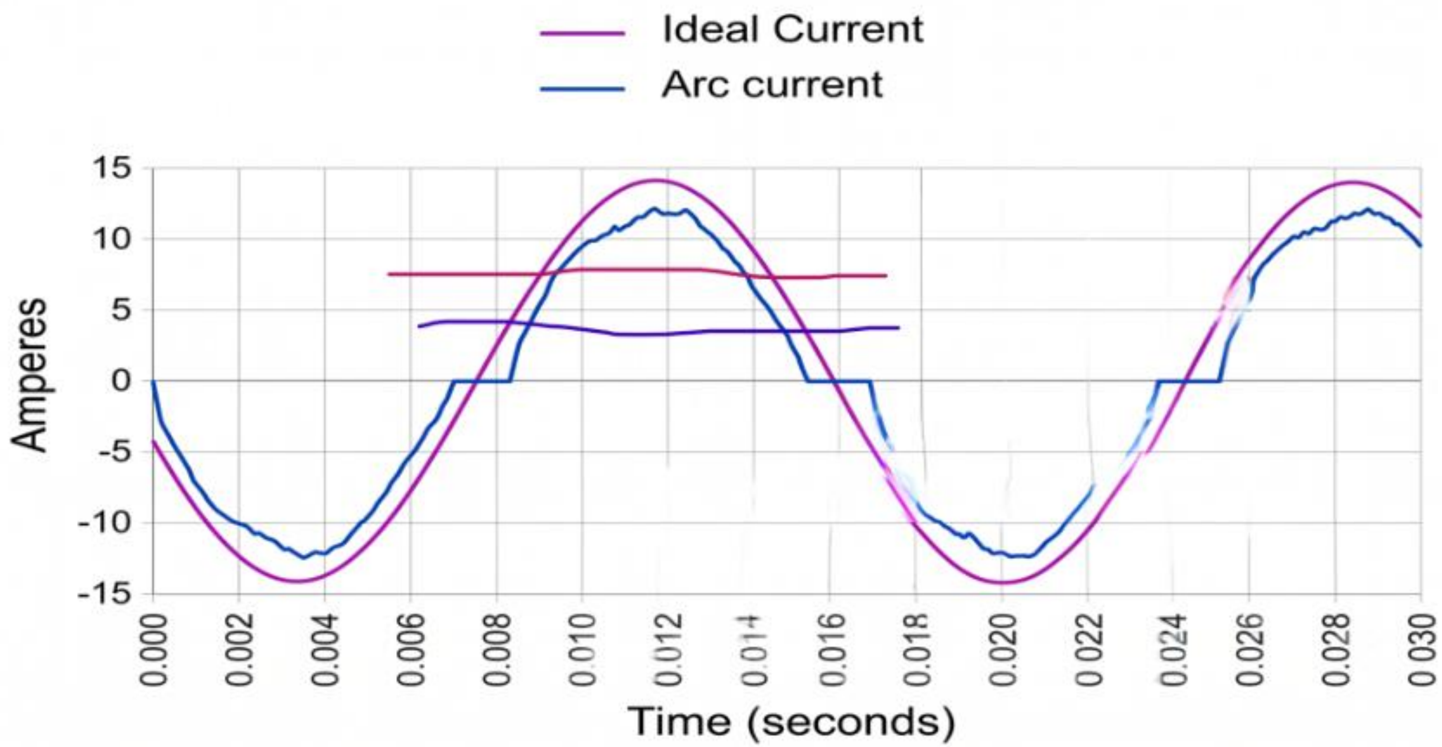
$$\frac{di}{dt}[n] \approx f_s \cdot (i[n] - i[n-1])$$





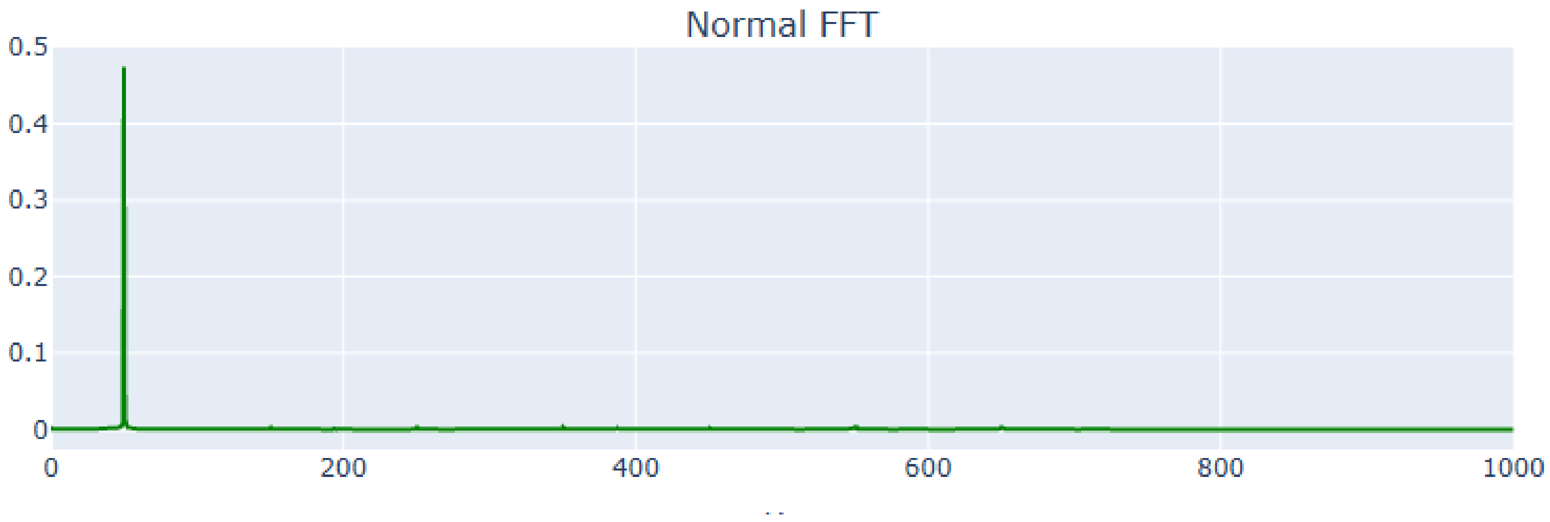
RMS Level

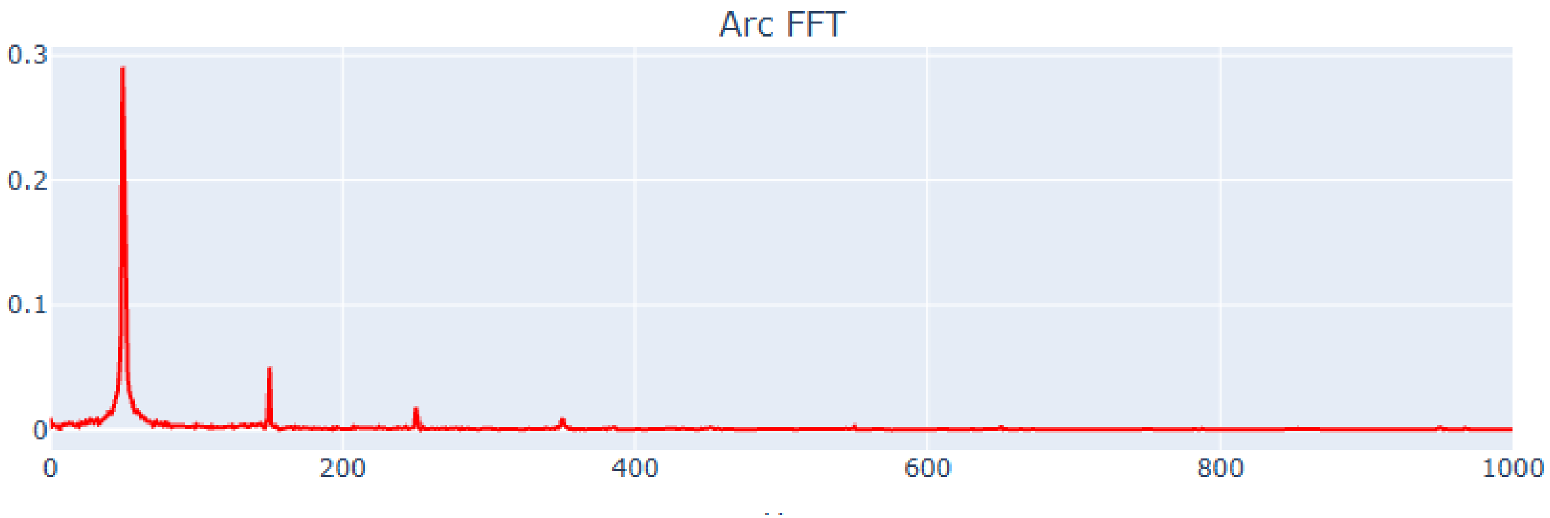
$$I_{RMS}[n] = \sqrt{\frac{1}{W}\sum_{k=n-\frac{W}{2}}^{n+\frac{W}{2}} i^2\,[k]}$$



Results

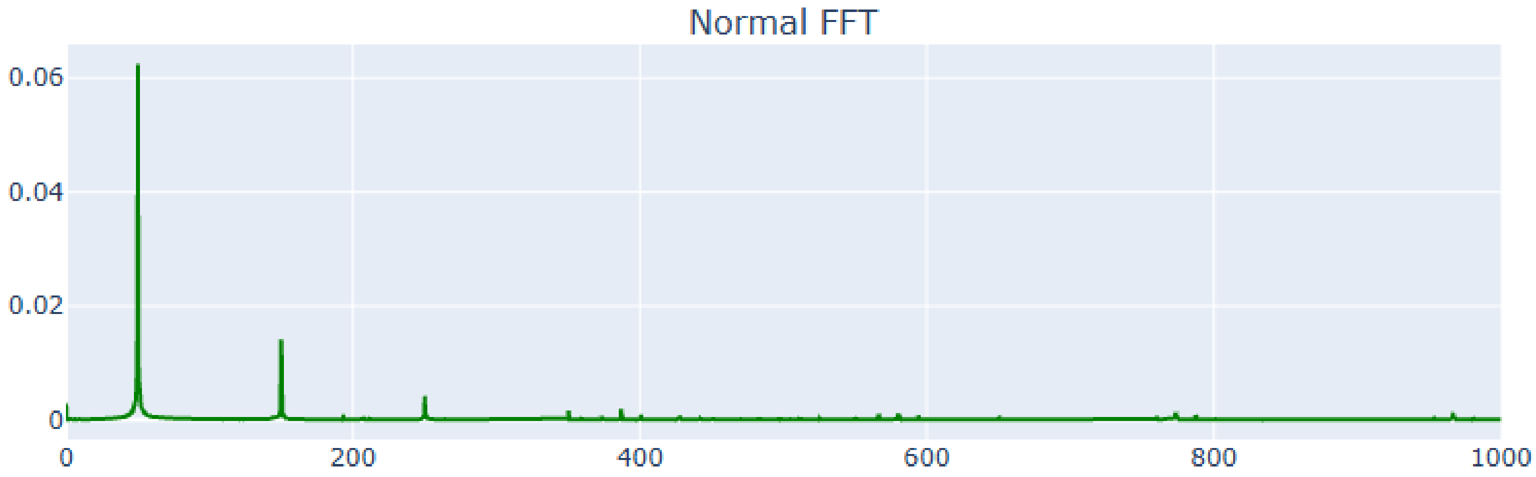
- **FFT (Resistive Load)**

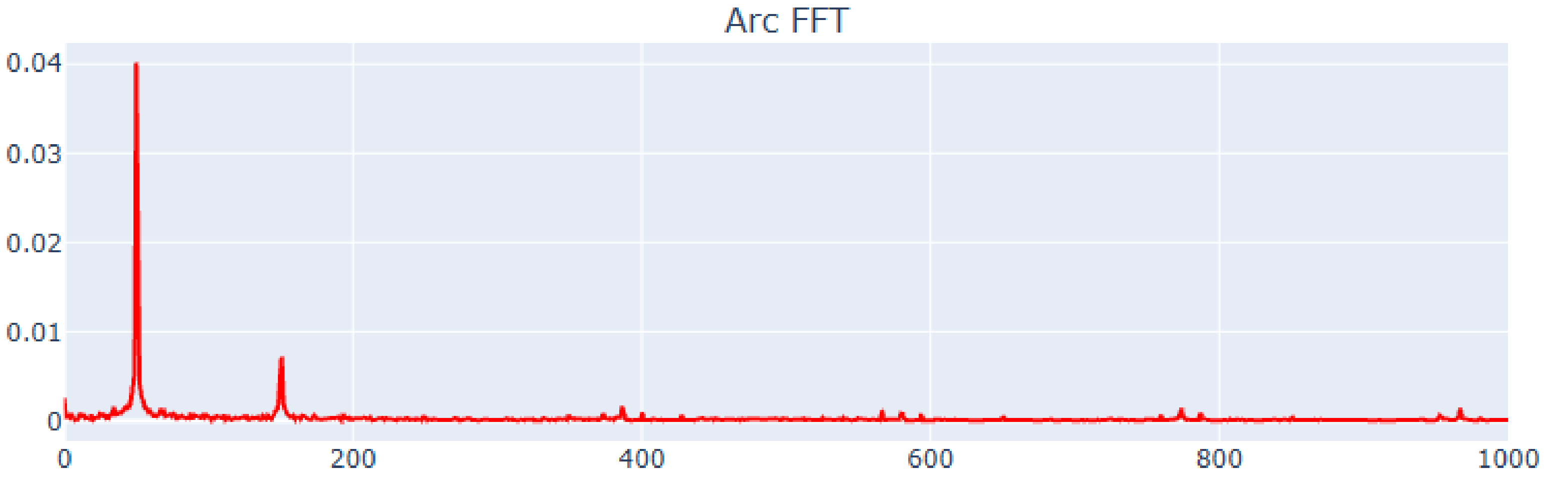




Results

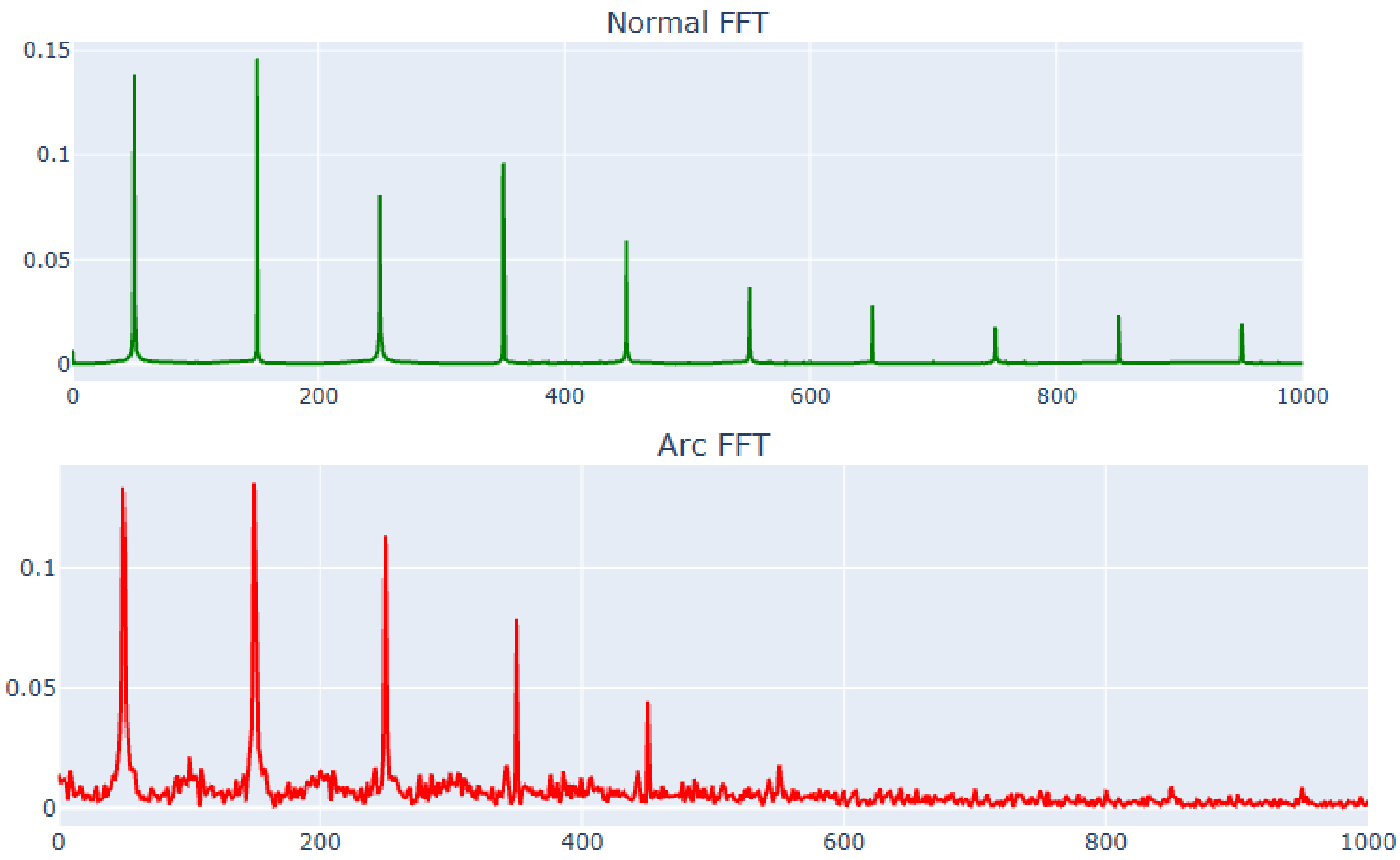
- **FFT (Inductive Load)**





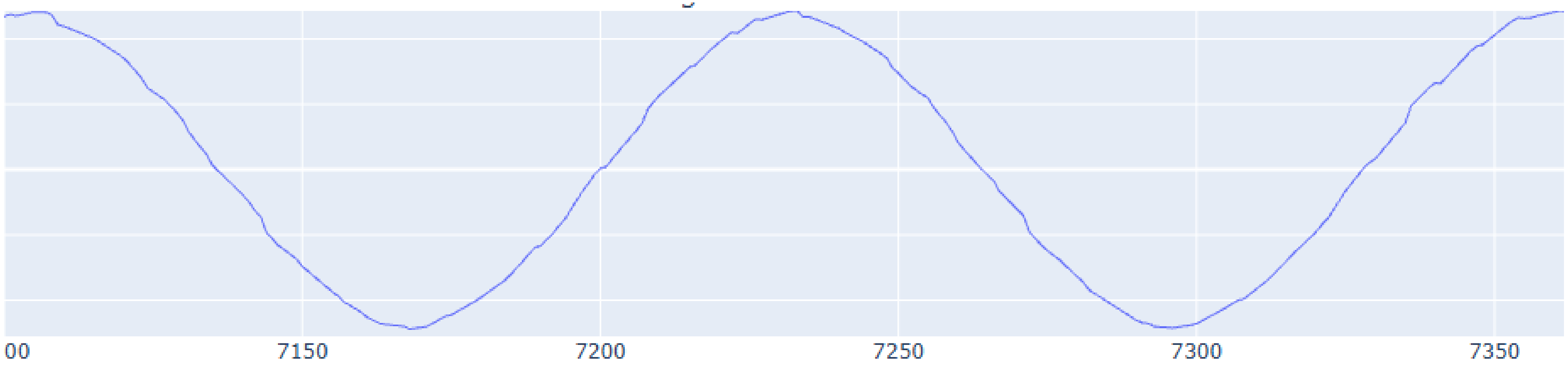
Results

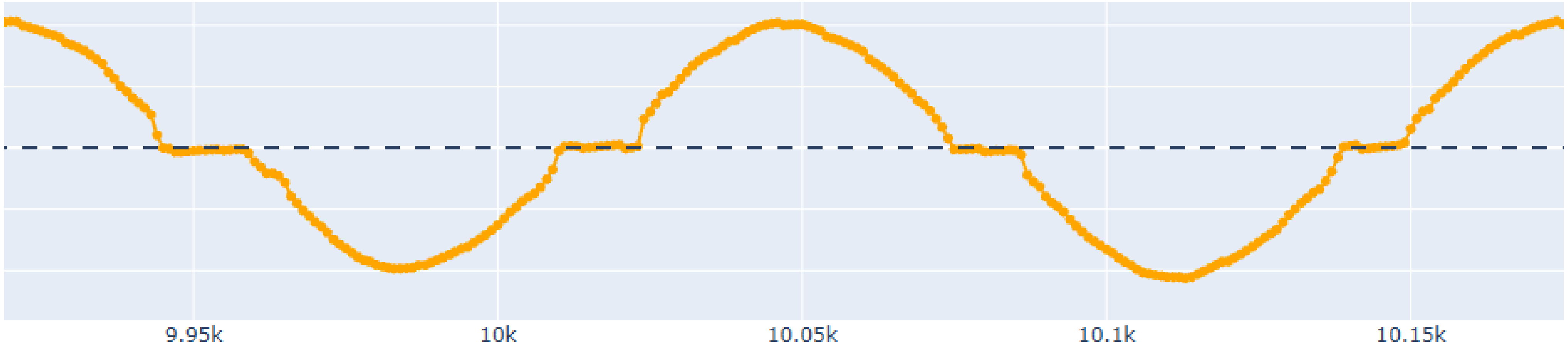
- **FFT(Switching Loads)**



Results

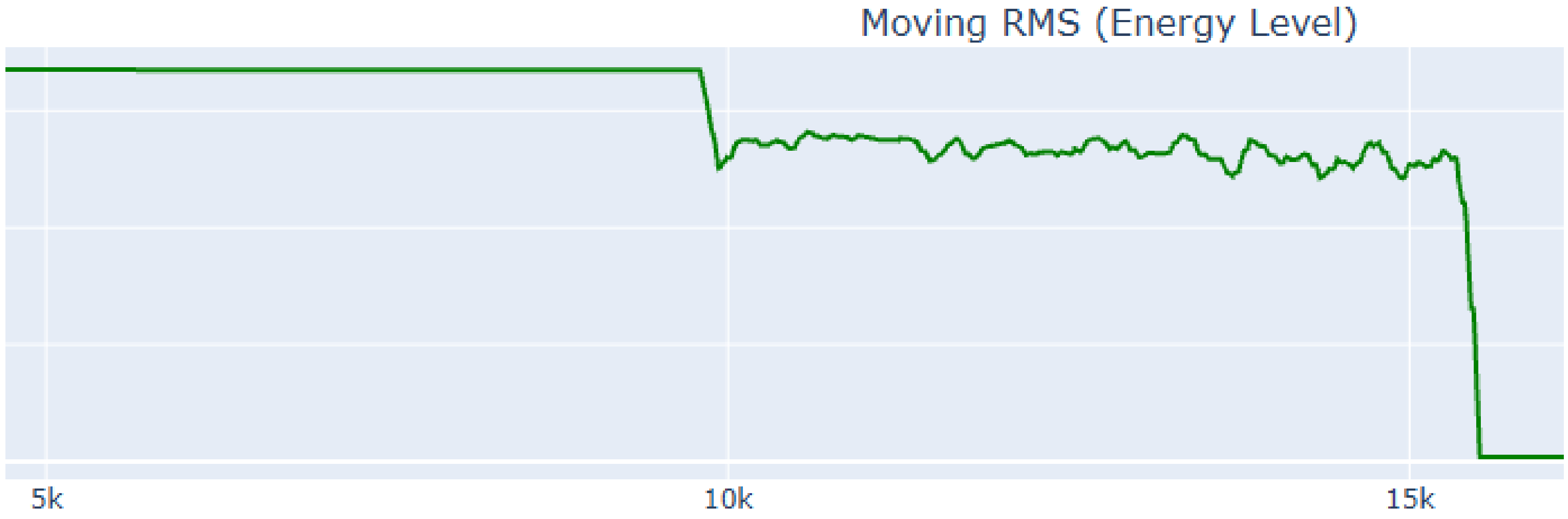
- **Zero Crossing**





Results

- **Moving RMS**



Results

- **Rate of change**